

TEJJAN ARORA

Monthly Progress Report February 2026

SAKEC TEAM

Exchange Functional PBE Convergence Test

Table 1: DFT Embedding with PBE Exchange: Energy Analysis

	Active	Reference DFT	Final Ground-State
Molecule	Space	Energy (Ha)	Energy (Ha)
H ₂	(2e, 2o)	−1.161905	−1.182605
H ₂ O	(2e, 6o)	−76.320335	−76.341891
O ₂	(8e, 6o)	−150.174613	−150.300681
O ₂	(2e, 6o)	−150.174613	−150.136870

Table 2: DFT Embedding with PBE Exchange: Convergence Analysis

	Active	Converged at	Energy	Wall
Molecule	Space	Iteration	Tolerance (Ha)	Time(sec)
H ₂	(2e, 2o)	9	$7.669e - 08$	61
H ₂ O	(2e, 6o)	9	$2.321e - 07$	1,081
O ₂	(8e, 6o)	27	$5.873e - 07$	9,217
O ₂	(2e, 6o)	4	$3.599e - 07$	141

Exchange Functional B3LYP Convergence Test

Table 3: B3LYP Exchange: Energy Analysis (alpha_init = 0.6)

Molecule	Active Space	Reference DFT Energy (Ha)	Final Ground-State Energy (Ha)
H ₂	(2e, 2o)	-1.175477	-1.190061620
H ₂ O	(2e, 6o)	-76.407023	-76.425277311

Table 4: B3LYP Exchange: Convergence Analysis (alpha_init = 0.6)

Molecule	Active Space	Converged at Iteration	Energy Tolerance (Ha)	Wall Time(sec)
H ₂	(2e, 2o)	9	$6.683e - 07$	61
H ₂ O	(2e, 6o)	9	$9.587e - 07$	1,115

Next Step : Test for Molecule O₂

GPU Acceleration: Progress & Roadblocks

Classical DFT Phase: Built a custom PySCF driver to intercept Qiskit Nature, forcing the initial reference calculation (gpu4pyscf.RKS) to execute natively and successfully on the GPU. Integrated CuPy array casting (**cp.asarray**) for the density matrix to force native GPU execution of the energy_tot and get_fock evaluations during the iterative loop.

Quantum VQE Phase: Integrated **AerEstimator** to handle the active space evaluation after transferring the mapped matrices from the custom driver.

Issue: Default Qiskit Nature workflows rely on NumPy arrays, which forces severe I/O latency due to constant memory transfers between the host (CPU) and device (GPU) during the embedding loops.

Success: The custom driver successfully converges the classical DFT on the GPU (**-75.177 Ha**) and successfully completes the Qiskit Mapper step.

Roadblock: The pipeline crashes during the hybrid handoff to the quantum simulator. The AerEstimator VQE execution aborts on the first iteration: **[CRITICAL ERROR] VQE failed on iteration 1: 'The primitive job failed!'**

Next Steps:

Debug Primitive Failure: Investigate the AerEstimator crash to determine if the matrix shapes passed from the custom driver are causing a timeout or incompatibility within the Qiskit Aer backend.

Complete GPU Isolation: Transition the remaining intermediate integral handling from NumPy to **CuPy** to ensure memory remains persistent on the GPU, avoiding the current CPU handoff step entirely.

Thank you

SAKEC TEAM